Intelligence Infrastructure: Scaling AI for the Frontier Era

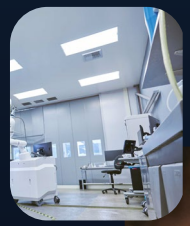
Reynold D’Sa



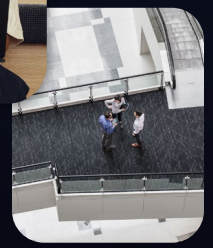
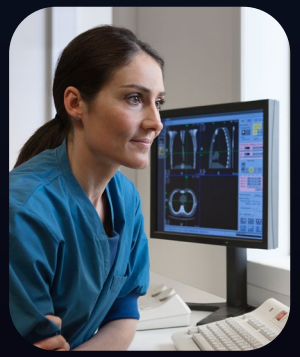
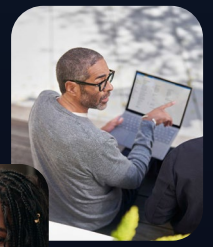
OCP GLOBAL SUMMIT

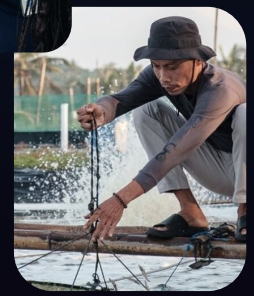
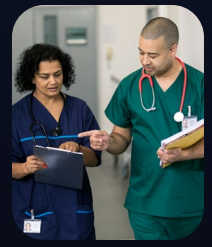
October 12-15, 2026
San Jose, California

Building the future — Frontier Firm





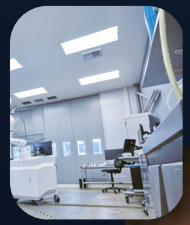




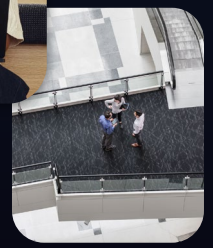
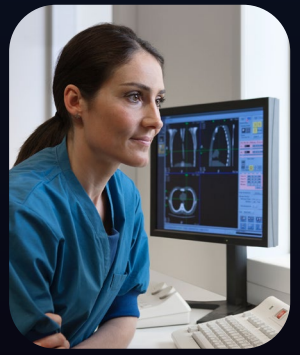
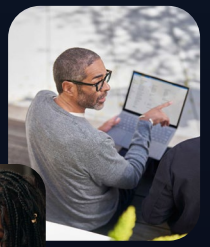


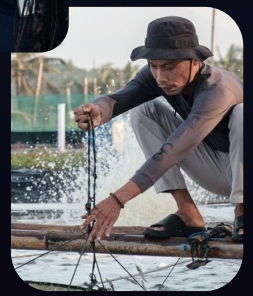
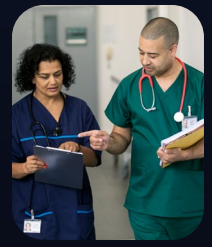

Building the future — Frontier Firm

Human capital + Token capital





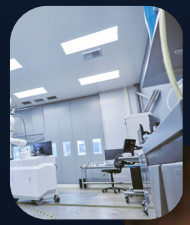




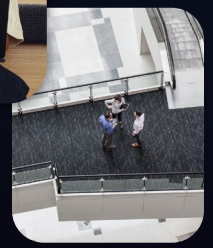
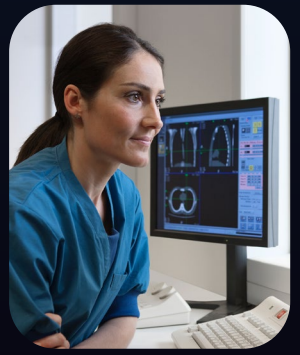
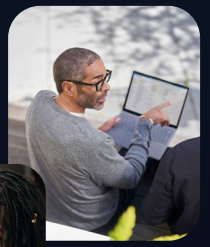


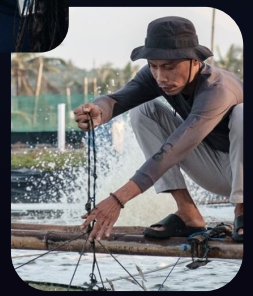
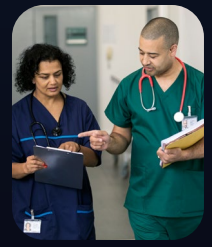

Building the future — Frontier Firm

Human capital + Token capital



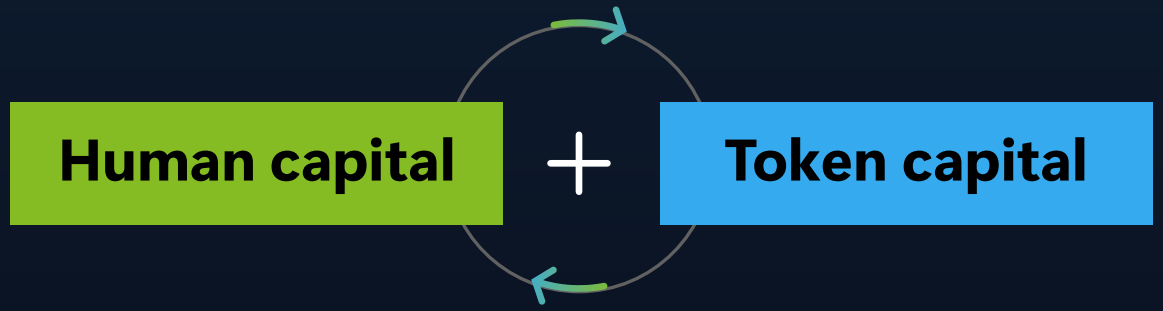


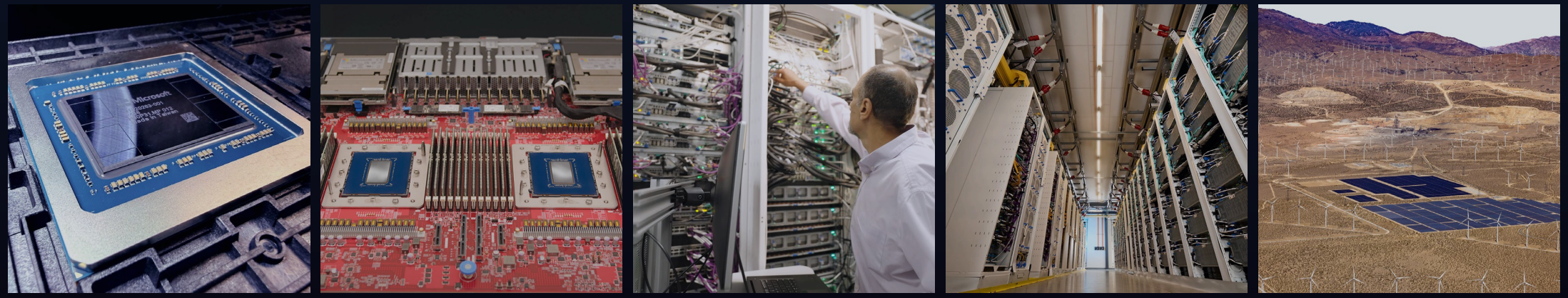





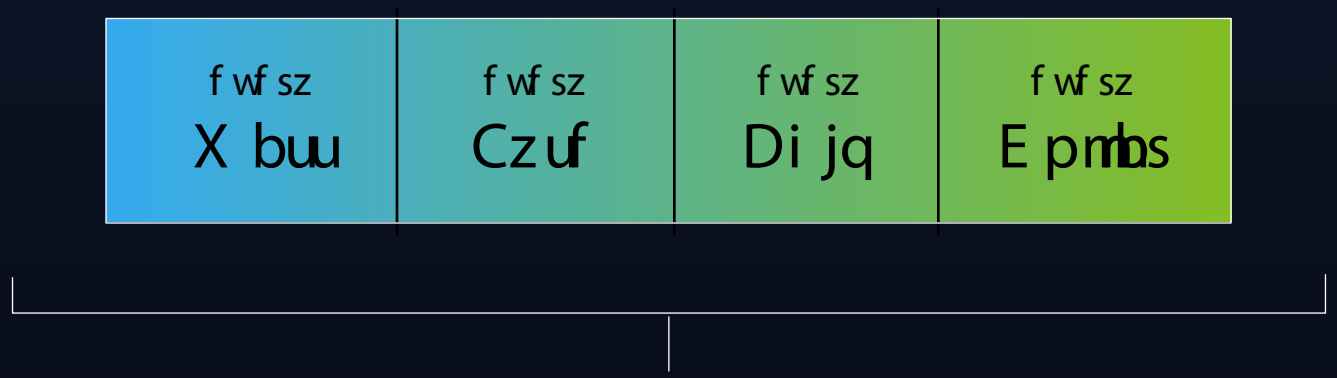

Building the future — Frontier Firm



Frontier infrastructure

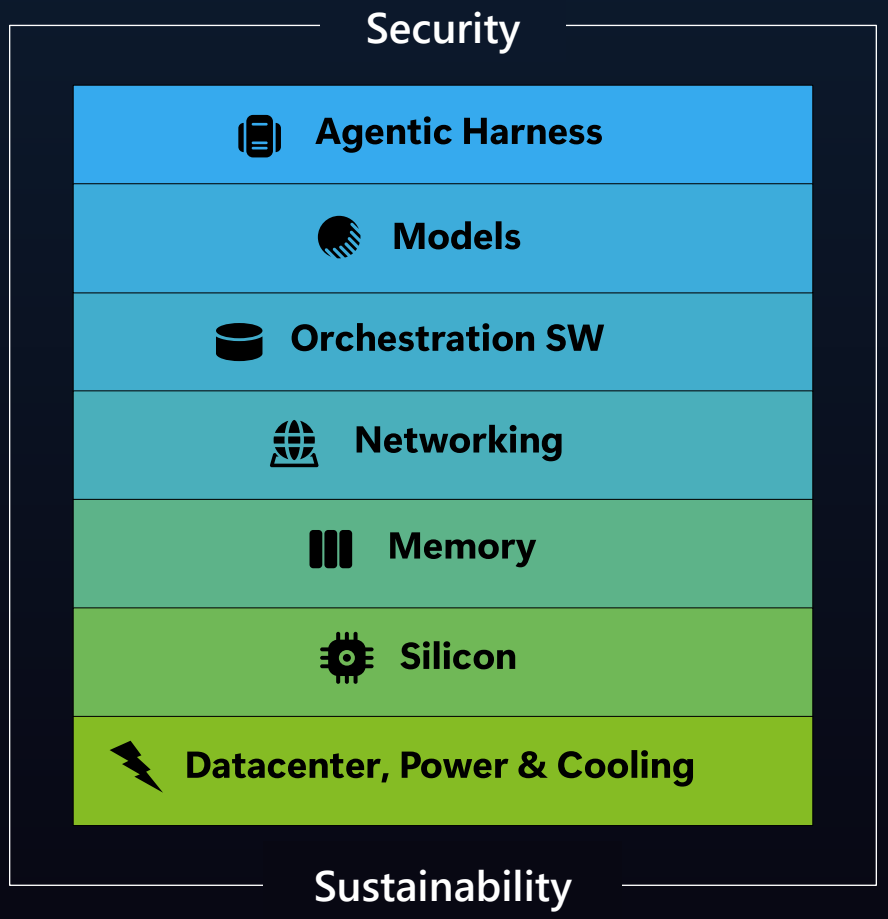


The yield imperative



Producing **useful** intelligence

Co-design across the stack



Co-design across the stack

Security

- Agentic Harness
- Models
- Orchestration SW
- Networking
- Memory
- Silicon
- Datacenter, Power & Cooling

Sustainability









From isolated AI experiments to AI-driven operations

Azure Maia: An Inference-First AI System



Maia 300

Advancing the Inference Efficiency Frontier

Up to 7× generation-over-generation inference performance

40% more tokens/MW vs rest of fleet when deployed

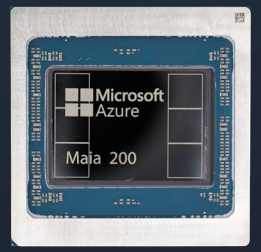
Scaling to Multi-GW Deployments

Starting in 2027

Purpose-Built Azure Silicon

End-to-end cloud-scale optimization

Maia 200



Up to 70%
More tokens/W vs rest of fleet

Scaling Globally

Cobalt 200

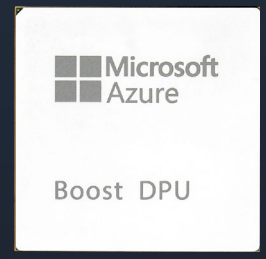


50%
Better performance gen-over-gen

33%
Lower agent call latency

23%
Higher throughput on agentic workloads

Boost DPU



4x
More performance

3x
Less power



Azure Boost

Offloaded host management



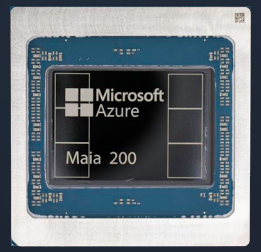
Azure Integrated HSM

In-house security chip in all new datacenter servers

Purpose-Built Azure Silicon

End-to-end cloud-scale optimization

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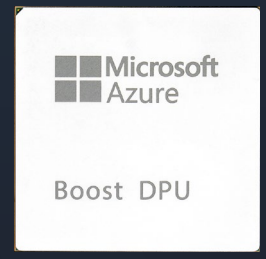


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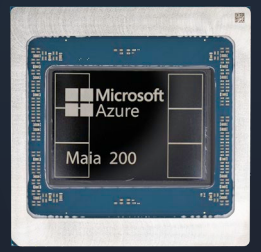
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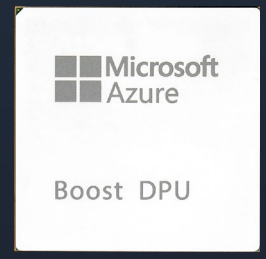


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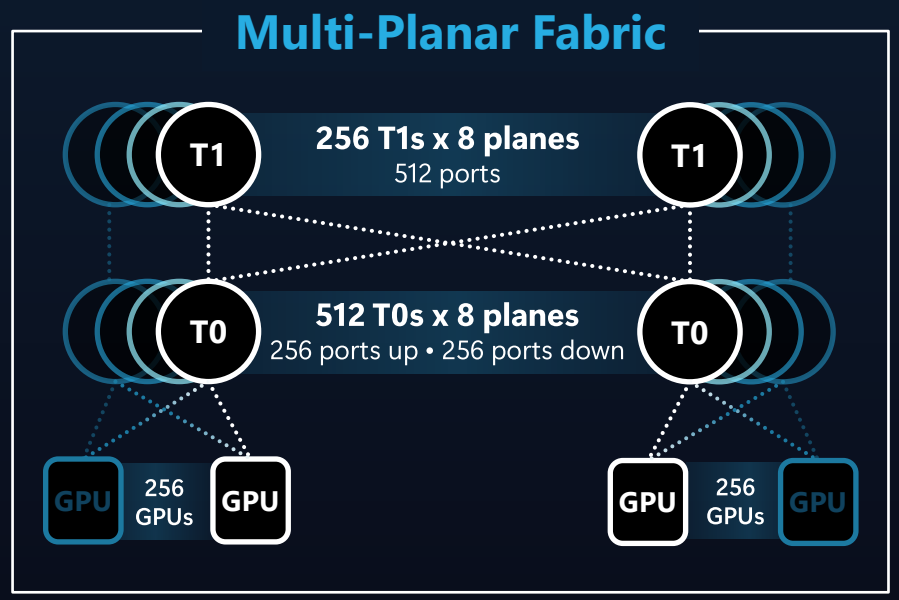


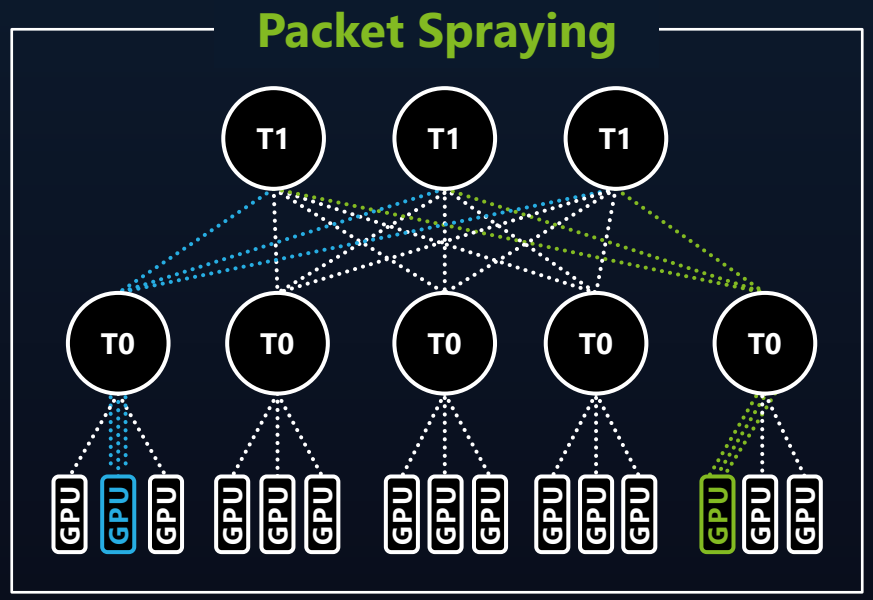
Azure Integrated HSM

In-house security chip in all new datacenter servers


Networking delivers the intelligence

Multipath Reliable Connection





- Frontier-scale
- Predictable & resilient
- Robust by design
- Open & switch-agnostic
- Routes around failures

Source: https://openai.com/index/mrc-supercomputer-networking

Microsoft AMD BROADCOM intel NVIDIA OpenAI

New AI fabrics & open infrastructure

Creating an Open Optical Specification for AI Scale Up

OCI-MSA

AMD · BROADCOM · NVIDIA · Meta · Microsoft · OpenAI



Building AI Infrastructure on Open Networks

OPEN Compute Project®

FTVO

New AI fabrics & open infrastructure

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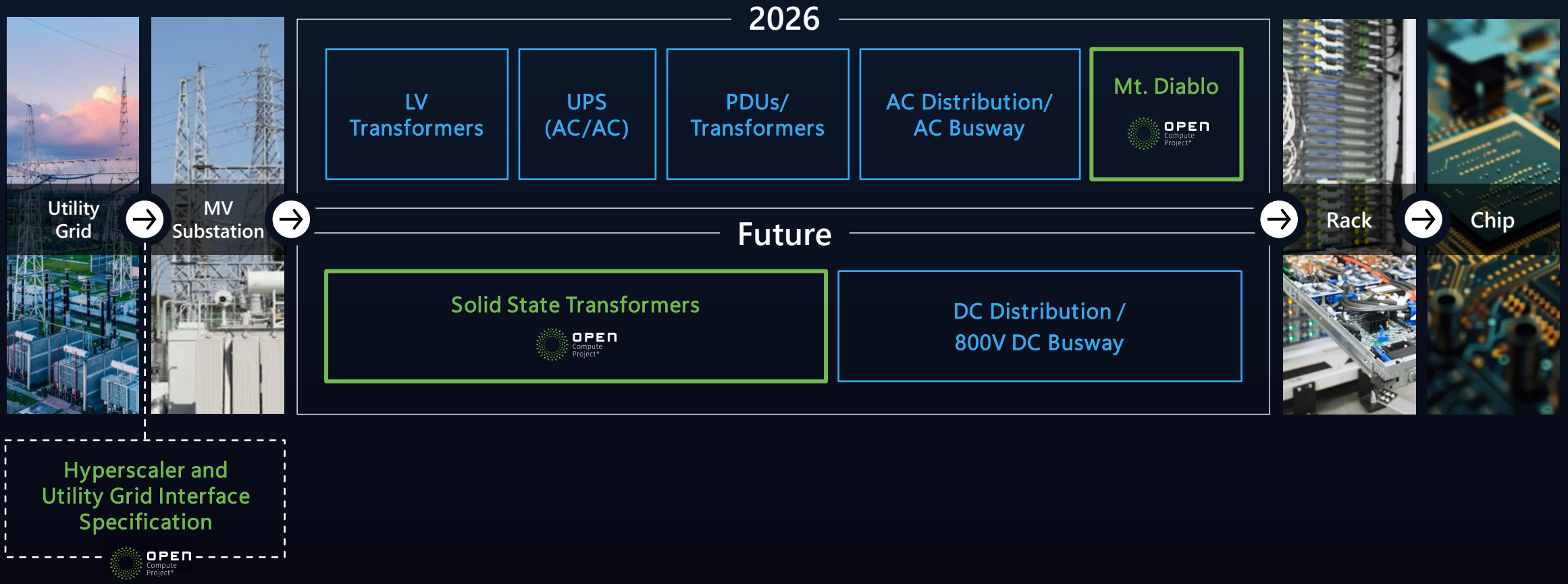


Building AI Infrastructure on Open Networks

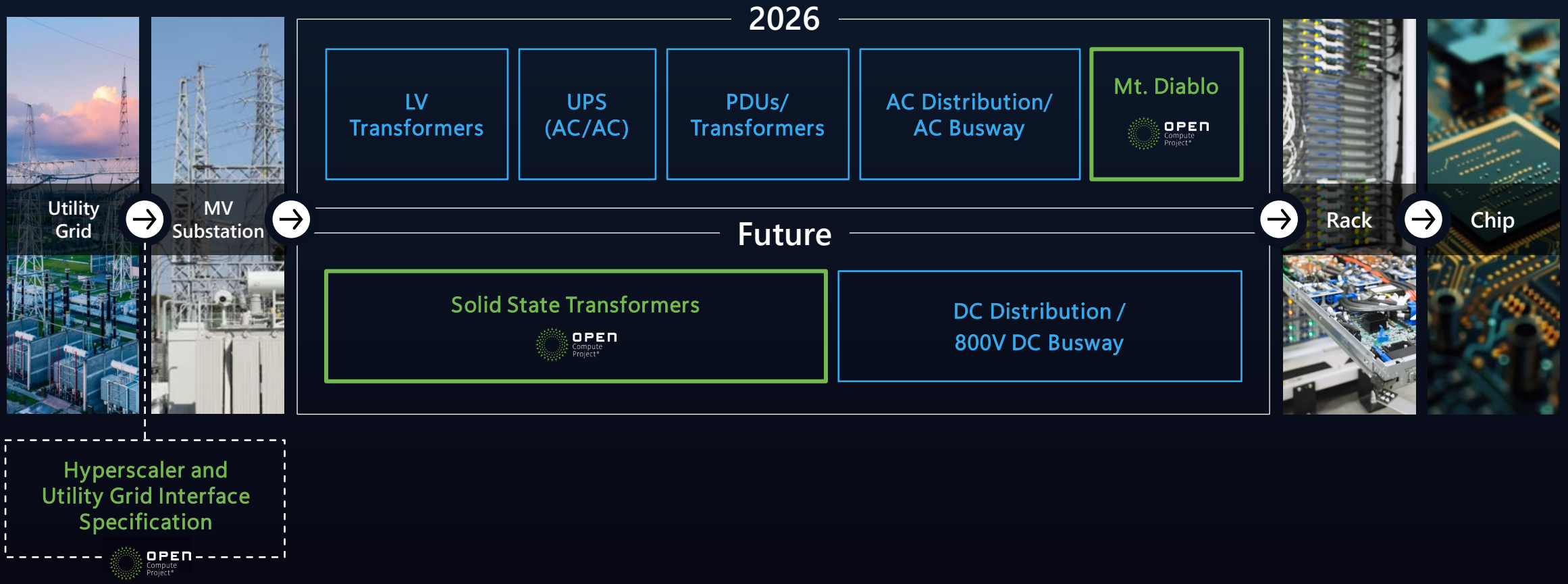
OPEN Compute Project®

FTVO

Re-thinking power from grid to chip



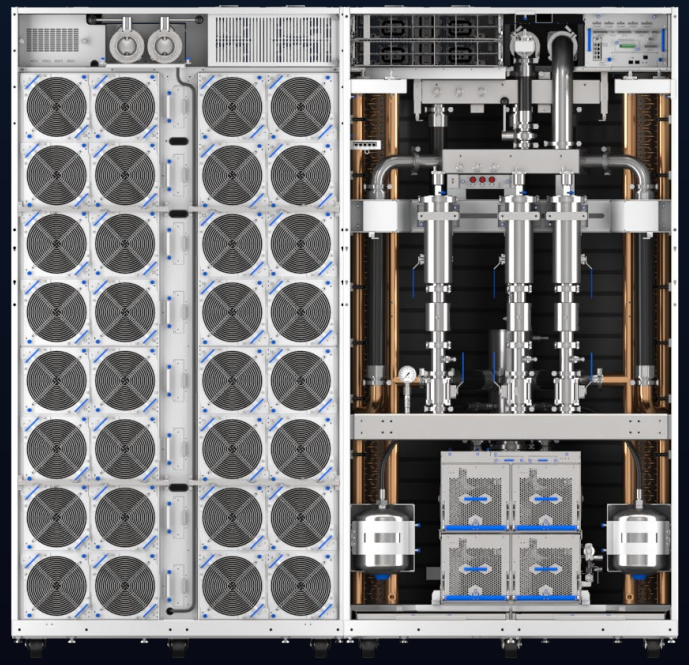
Re-thinking power from grid to chip



Cooling is a first-class design consideration

Mt. Atlas

~2x
Cooling capacity in the same footprint



Zonal Cooling

Liquid Cooled Heat rejection
20°C to 40°C water supply

Coolant Distribution Unit
Up to 45°C liquid supply

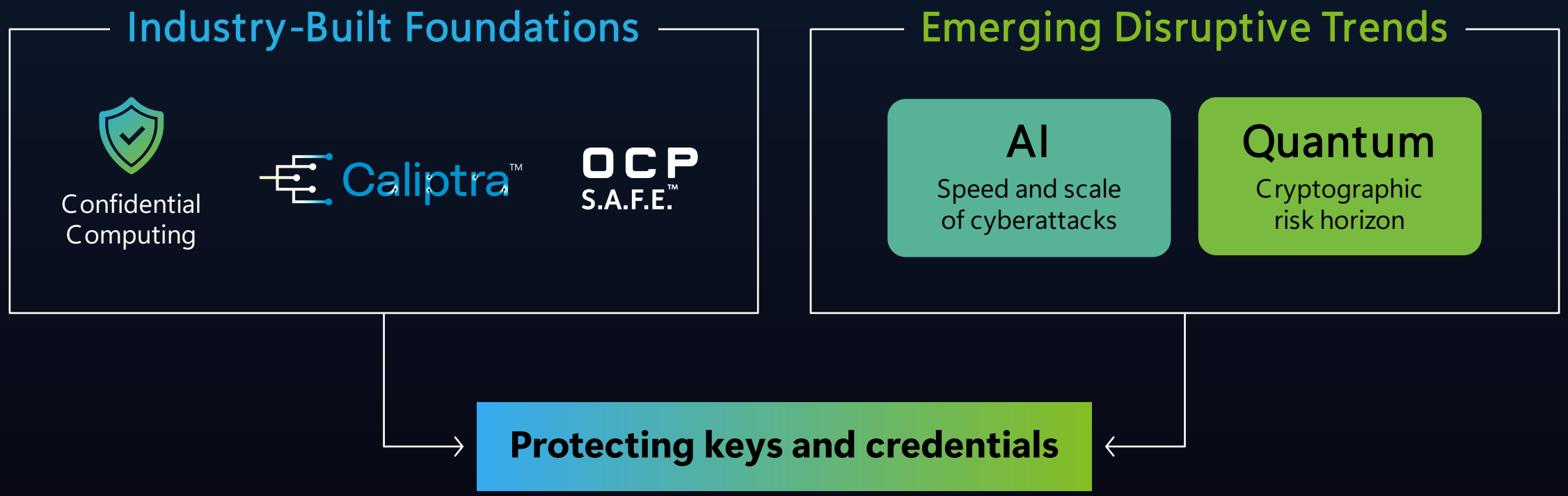
GPU racks

Air Cooled Heat rejection
20°C water supply

Air Handling Unit
30°C air supply

Air-cooled racks

Open foundations for the next era of hardware security



Tessera Integrated HSM

Keys never exposed in memory

Virtual Machine

Code

Memory

Guest OS

Host OS

Hypervisor

Integrated HSM



Keys isolated, not leaving Hardware Security Module (HSM) boundary

OPEN Compute Project

Microsoft

MARVELL

ORACLE

Fleet reliability at scale

Standardize

- RAS
- Debug & Diagnostics
- F/W Updates
- Manageability



Fleet Deployment

- Driving Compliance
- Silicon & Platform Intercepts

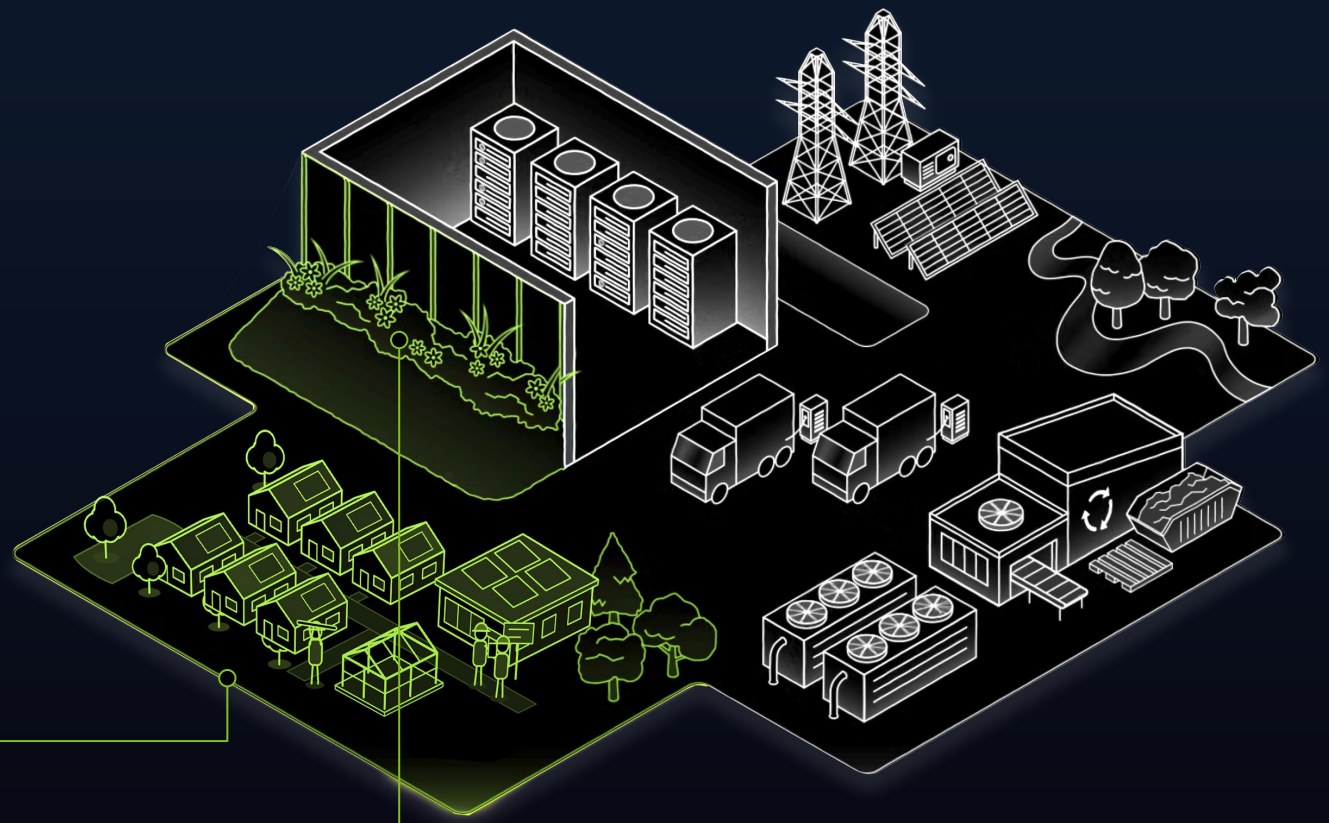
Self-managed fleet

Prevent, predict and AI-driven fleet actions.
Higher Nodes-In-Service with reduced manual effort.

Scaling AI sustainably

Zero Waste	Carbon Negative	Water Positive
		

Scaling AI sustainably



Supporting community ecosystem restoration

Designing for local conditions and ecosystems

Scaling AI sustainably

Advancing low-carbon construction materials



Timber framing | Low-carbon concrete | Green steel

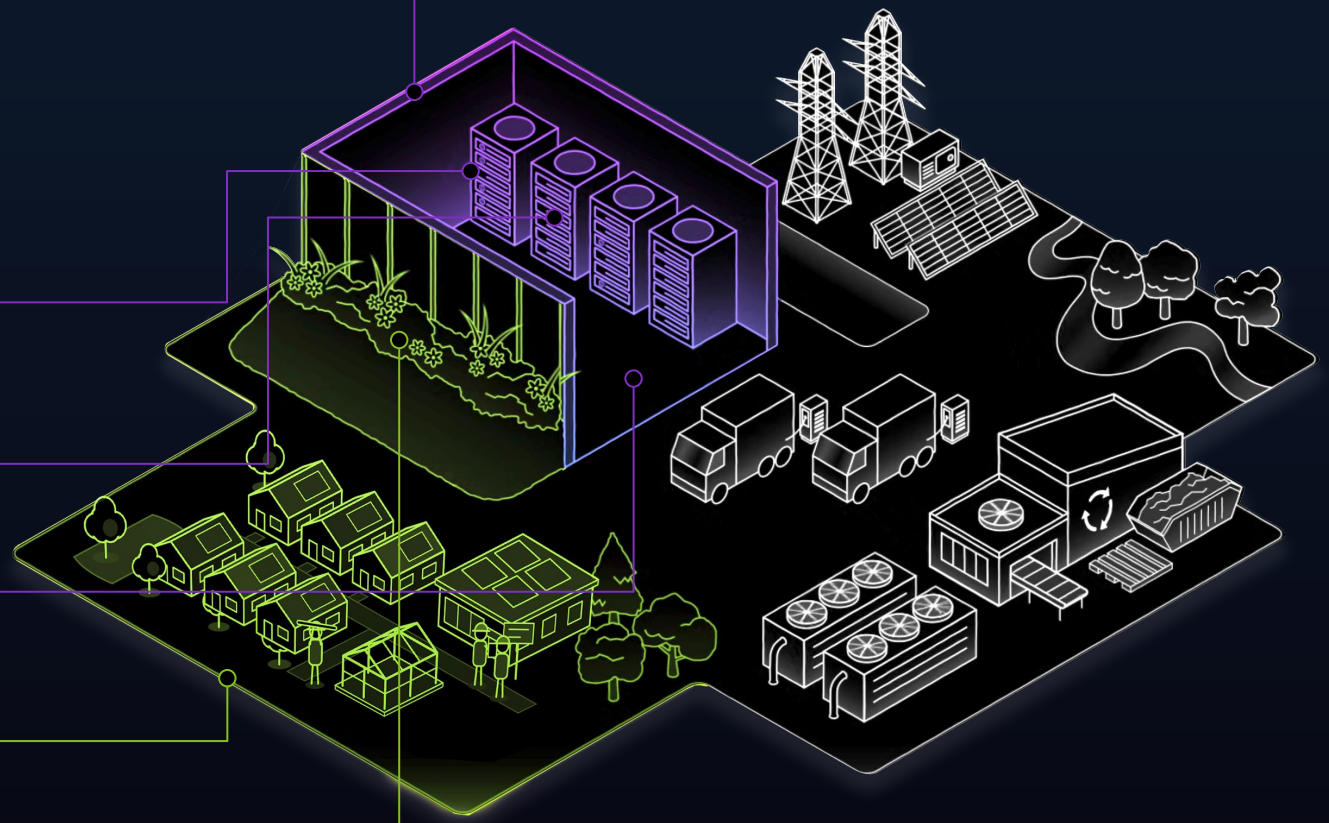
Reducing water consumption for cooling

Improving power and compute efficiency

Certifying datacenter performance through LEED

Supporting community ecosystem restoration

Designing for local conditions and ecosystems



Scaling AI sustainably

Advancing low-carbon construction materials



Timber framing | Low-carbon concrete | Green steel

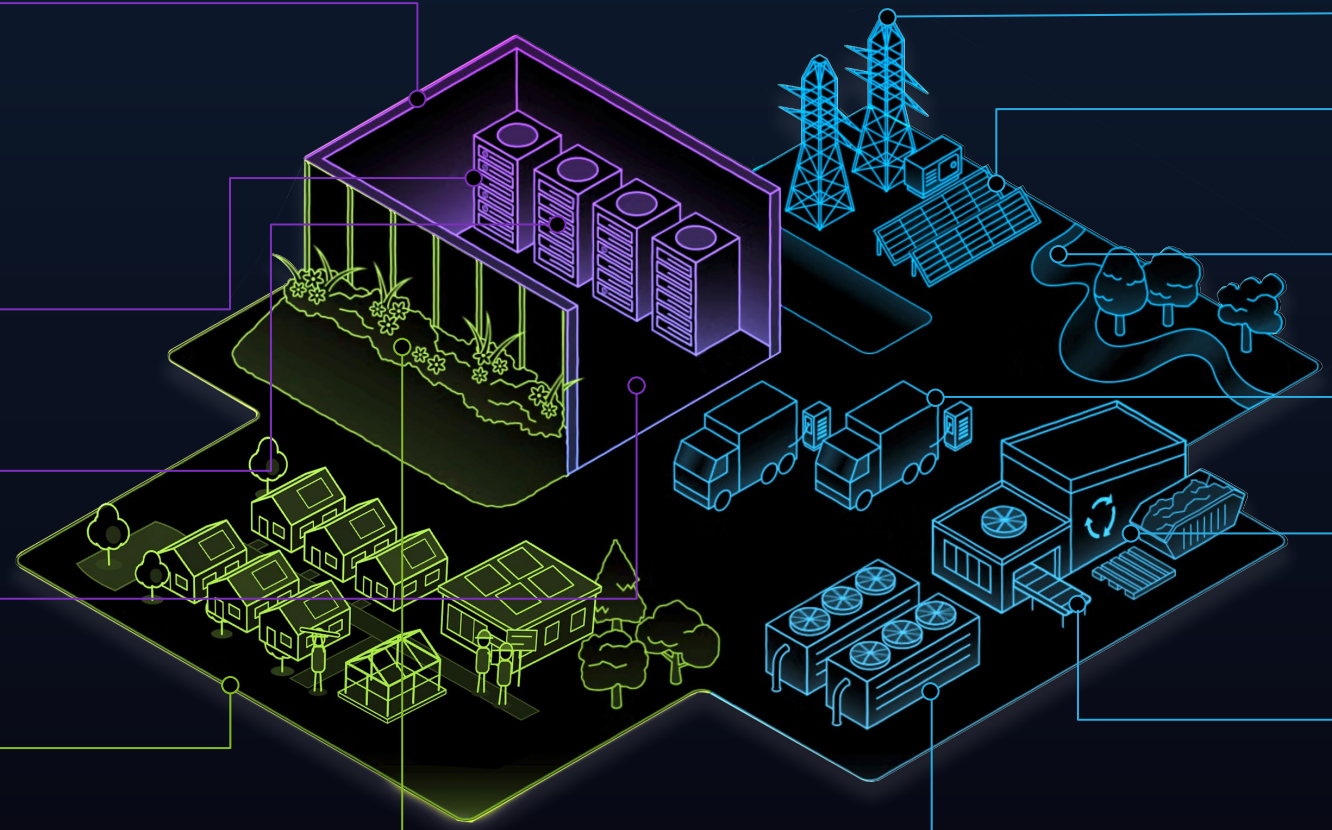
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Advancing energy markets

Investing in clean power and fuels

Strengthening water replenishment

Reducing emissions in our logistics network

Reducing waste in datacenter construction and operations

Improving circularity of cloud hardware

Repurposing waste heat

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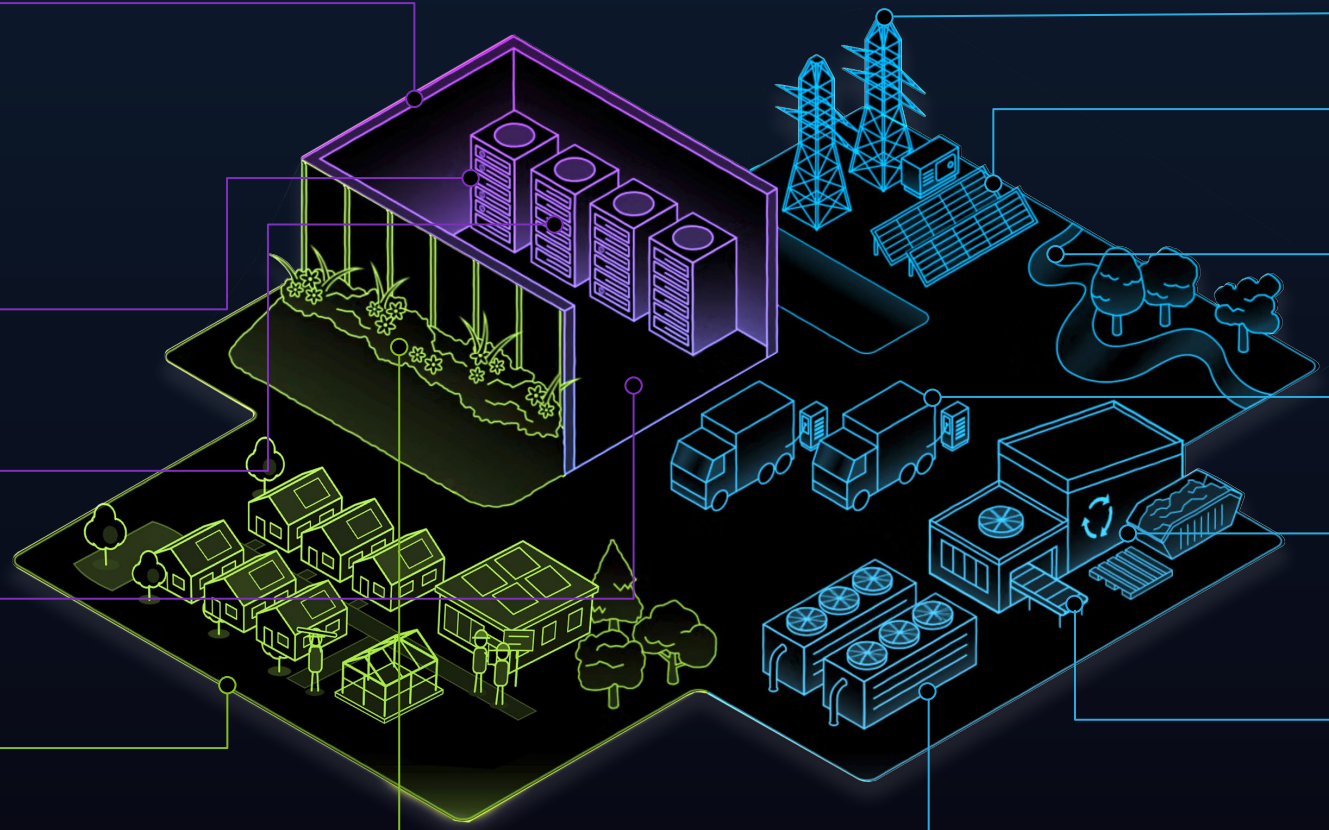
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Repurposing waste heat

From frontier infrastructure to a frontier firm

Frontier firm

Built on the platform

Enterprise platform

Choice across models & silicon

Frontier Infrastructure

Full Stack co-design for enhanced efficiency – the Yield Imperative.

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Frontier firm

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Continued Partnership & Collaboration

Thank You

OPEN Compute Project®

Microsoft

Microsoft at OCP Global Summit 2026